Supplementary Material: Auxiliary Algorithms for "Steering Through the Trees: Physical Lie Vector Field Evasion of Tree Ensembles in Network Intrusion Detection"

Quang-Vinh Dang

S1. Proofs

Proposition 1 and Theorem 1 are stated in Section 4.2 of the main paper; the notation is that of Section 4.1.

Proof of Proposition 1. We evaluate the vector bracket $[X, Y] = J_Y X - J_X Y$ coordinate-wise across all six distinct generator pairs:

1. **Non-Derived Coordinates $\{0, 1, 2, 3, 4, 5, 6, 7, 11\}$:** For these coordinates, the components of all four generators are either constants $(0, 1, L_0, H_0)$ or depend solely on their own coordinate (e.g., $\vec{v}_{\text{win}}$ depends only on x_7). Since no generator modifies an independent variable on which another generator depends, the Jacobian products vanish identically: $(J_Y X)_k = 0$ and $(J_X Y)_k = 0$ for all $k \notin \{8, 9, 10\}$.

2. **Temporal Dilation and Payload Padding ($X = \vec{v}_{\textbf{time}}, Y = \vec{v}_{\textbf{pad}}$):** X has non-zero entries in rows $6, 8, 9$ ($X_6 = 1, X_8 = -x_8/x_6, X_9 = -x_9/x_6$); Y has non-zero entries in rows $4, 8, 10$ ($Y_4 = 1, Y_8 = 10^6/x_6, Y_{10} = 1/x_2$). In row 8:

$$(J_Y X)_8 = \frac{\partial Y_8}{\partial x_6} X_6 = -\frac{10^6}{x_6^2} \cdot 1 = -\frac{10^6}{x_6^2} \tag{1}$$

$$(J_X Y)_8 = \frac{\partial X_8}{\partial x_4} Y_4 + \frac{\partial X_8}{\partial x_8} Y_8 = 0 + \left(-\frac{1}{x_6}\right)\left(\frac{10^6}{x_6}\right) = -\frac{10^6}{x_6^2} \tag{2}$$

   Subtracting yields $[X, Y]_8 = -\frac{10^6}{x_6^2} - \left(-\frac{10^6}{x_6^2}\right) \equiv 0$. In rows 9 and 10, all corresponding partial derivatives vanish, so $[X, Y] = \mathbf{0}$.

3. **Temporal Dilation and Chaff Injection ($X = \vec{v}_{\textbf{time}}, Z = \vec{v}_{\textbf{chaff}}$):** Z has entries $Z_2 = 1, Z_4 = L_0, Z_8 = L_0 \times 10^6/x_6, Z_9 = 10^6/x_6, Z_{10} = (L_0 - x_{10})/x_2, Z_{11} = H_0$. In row 8: $(J_Z X)_8 = -L_0 \times 10^6/x_6^2 = (J_X Z)_8$. In row 9: $(J_Z X)_9 = -10^6/x_6^2 = (J_X Z)_9$. In row 10: $(J_Z X)_{10} = 0 = (J_X Z)_{10}$. Thus $[X, Z] = \mathbf{0}$.

4. **Payload Padding and Chaff Injection ($Y = \vec{v}_{\mathbf{pad}}, Z = \vec{v}_{\mathbf{chaff}}$):** In ambient $\mathbb{R}^{12}$ coordinates, $Y_{10}(x) = 1/x_2$ and $Z_{10}(x) = (L_0 - x_{10})/x_2$. Evaluating the ambient cross-Jacobian products in row 10:

$$(J_Z Y)_{10} = \sum_{k=0}^{11} \frac{\partial Z_{10}}{\partial x_k} Y_k = \frac{\partial Z_{10}}{\partial x_{10}} Y_{10} = \left(-\frac{1}{x_2}\right)\left(\frac{1}{x_2}\right) = -\frac{1}{x_2^2} \tag{3}$$

$$(J_Y Z)_{10} = \sum_{k=0}^{11} \frac{\partial Y_{10}}{\partial x_k} Z_k = \frac{\partial Y_{10}}{\partial x_2} Z_2 = \left(-\frac{1}{x_2^2}\right) \cdot 1 = -\frac{1}{x_2^2} \tag{4}$$

Subtracting gives $[Y, Z]_{10} = (J_Z Y)_{10} - (J_Y Z)_{10} = -\frac{1}{x_2^2} - \left(-\frac{1}{x_2^2}\right) \equiv 0$. In rows 8 and 9, all entries vanish identically because $Y_6 = Z_6 = 0$ while the rate entries Y_8, Z_8, Z_9 depend spatially only on duration x_6, ensuring $(J_Z Y)_k = (J_Y Z)_k = 0$ for $k \in \{8, 9\}$.

5. **Pairs Involving Window Morphing $\vec{v}_{\mathbf{win}}$:** Because $\vec{v}_{\text{win}}$ acts exclusively on x_7, which is neither read nor modified by $\vec{v}_{\text{time}}, \vec{v}_{\text{pad}}, \vec{v}_{\text{chaff}}$, all cross-Jacobians vanish identically: $[\vec{v}_k, \vec{v}_{\text{win}}] = \mathbf{0}$ for all $k \in \{1, 2, 3\}$.

Hence, $[\vec{v}_i, \vec{v}_j] = \mathbf{0}$ identically for all pairs $i, j \in \{1, 2, 3, 4\}$. □

Corollary S1 (Frobenius Integrability and Commuting Flows)**.** *On the regular open subdomain where the four generators are linearly independent (specifically where $x_6 > 0$, $x_2 \geq 1$, and $x_7 \neq W_{target}$ for TCP), the distribution $\Delta = \text{span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3, \vec{v}_4\}$ has constant rank 4. Because $[\vec{v}_i, \vec{v}_j] \equiv \mathbf{0}$, Δ is involutive and completely integrable by Frobenius' Theorem. At points where $x_7 = W_{target}$, $\vec{v}_4 = \mathbf{0}$ and Δ smoothly reduces to the 3-dimensional involutive distribution spanned by $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$. Furthermore, the generated continuous flow maps commute:*

$$\varphi_i^{s_1} \circ \varphi_j^{s_2} = \varphi_j^{s_2} \circ \varphi_i^{s_1}, \quad \forall s_1, s_2 \geq 0 \tag{5}$$

forming an Abelian local Lie semigroup action (commuting local flows $\varphi_i^{s_1} \circ \varphi_j^{s_2} = \varphi_j^{s_2} \circ \varphi_i^{s_1}$ for all $s_1, s_2 \geq 0$ on the interior $\widetilde{\mathcal{M}}_{flow}^{\circ}$). Consequently, on the manifold interior where bounding constraints remain slack, the terminal flow state depends solely on the net integrated control amounts $(s_1, s_2, s_3, s_4) = \int_0^T u(t)dt$ and is strictly invariant to the sequential order in which infinitesimal operations are executed. When active boundary hypersurfaces are encountered (such as the MTU envelope $g_1(x) = 1500x_2 - x_4 = 0$), control admissibility $\mathcal{U}(x)$ restricts admissible directions, requiring interleaving chaff packets ($u_3 > 0$) to expand total packet capacity before additional payload padding ($u_2 > 0$) can be absorbed.

Proof of Theorem 1. Let Ψ be the constraint residual defined in Section 4.1 of the main paper. Differentiating $\Psi_1(x(t))$ along the trajectory $\dot{x}(t) = \sum_{k=1}^{4} u_k(t)\vec{v}_k(x(t))$:

$$\begin{aligned}
\frac{d}{dt}\Psi_1(t) &= \dot{x}_8 - \frac{10^6(\dot{x}_4 + \dot{x}_5)}{x_6} + \frac{10^6(x_4 + x_5)\dot{x}_6}{x_6^2} \\
&= \left(-u_1\frac{x_8}{x_6} + u_2\frac{10^6}{x_6} + u_3\frac{L_0 \cdot 10^6}{x_6}\right) - \frac{10^6(u_2 + L_0 u_3)}{x_6} + \frac{10^6(x_4 + x_5)u_1}{x_6^2} \\
&= -\frac{u_1(t)}{x_6(t)}\left(x_8 - \frac{10^6(x_4 + x_5)}{x_6}\right) = -\frac{u_1(t)}{x_6(t)}\Psi_1(t)
\end{aligned} \tag{6}$$

Identical algebraic cancellation yields:

$$\frac{d}{dt}\Psi_2(t) = -\frac{u_1(t)}{x_6(t)}\Psi_2(t), \quad \frac{d}{dt}\Psi_3(t) = -\frac{u_3(t)}{x_2(t)}\Psi_3(t) \tag{7}$$

where the Ψ_3 identity follows by substituting $\dot{x}_{10} = (u_2 + u_3(L_0 - x_{10}))/x_2$, $\dot{x}_4 = u_2 + L_0 u_3$ and $\dot{x}_2 = u_3$ into $\dot{x}_{10} - \dot{x}_4/x_2 + x_4\dot{x}_2/x_2^2$. Hence, the constraint error vector $\Psi(t)$ satisfies the uncoupled homogeneous linear differential system $\dot{\Psi}(t) = A(t)\Psi(t)$, where $A(t) = \mathrm{diag}(-u_1(t)/x_6(t), -u_1(t)/x_6(t), -u_3(t)/x_2(t))$. Since $x_6(t) \geq x_6(0) > 0$ and $x_2(t) \geq x_2(0) \geq 1$, $A(t)$ is locally Lebesgue integrable ($A \in L^1_{\mathrm{loc}}$). Because $x_0 \in \widetilde{\mathcal{M}}_{\mathrm{flow}}$, the initial condition is $\Psi(0) = \mathbf{0}$. By the Carathéodory existence and uniqueness theorem for linear differential equations (Carathéodory, 1918), the unique absolutely continuous solution is $\Psi(t) \equiv \mathbf{0}$ for all $t \geq 0$.

Furthermore, examining the base physical coordinate trajectories across the boundary faces of $\widetilde{\mathcal{M}}_{\mathrm{flow}}$:

- $\dot{x}_0 = \dot{x}_1 = \dot{x}_3 = \dot{x}_5 \equiv 0$, satisfying strict immutability for $\mathcal{I}_{\mathrm{imm}}$.
- $\dot{x}_6 = u_1(t) \geq 0 \implies x_6(t) \geq x_6(0) > 0$, and $x_6(t) \leq 1.1 \times 10^8$ by the cap rule $u_1 = 0$ at the timeout boundary.
- $\dot{x}_2 = u_3(t) \geq 0 \implies x_2(t) \geq x_2(0) \geq 1$.
- **Total Bytes and Header Consistency ($x_4 \geq x_{11} + 20x_2$):** Chaff packets carry zero application payload; each chaff packet adds exactly 20 bytes of IPv4 header plus transport header H_0 ($H_0 = 20$ for minimal TCP, 32 with TSOPT, 8 for UDP), so its total packet size is $L_0 = 20 + H_0$. Differentiating the gap function $h(x) = x_4 - x_{11} - 20x_2$:

$$\begin{aligned} \dot{h}(x(t)) &= \dot{x}_4 - \dot{x}_{11} - 20\dot{x}_2 = (u_2 + L_0 u_3) - H_0 u_3 - 20u_3 \\ &= u_2(t) + (L_0 - H_0 - 20)u_3(t) = u_2(t) \geq 0 \end{aligned} \tag{8}$$

  Because $\dot{h}(t) = u_2(t) \geq 0$, the inequality $x_4(t) \geq x_{11}(t) + 20x_2(t)$ is monotonically non-decreasing along all Lie trajectories.

- **Transport Header Bounds ($H_{\min}x_2 \leq x_{11} \leq H_{\max}x_2$):** For TCP ($x_1 = 6$), $H_0 \in [20, 60]$, yielding $\dot{x}_{11} = H_0 u_3(t) \in [20, 60]\dot{x}_2(t)$, ensuring $20x_2(t) \leq x_{11}(t) \leq 60x_2(t)$ is forward-invariant; for UDP ($x_1 = 17$), $H_0 = 8$ strictly, maintaining $x_{11}(t) = 8x_2(t)$ identically.
- **Nagumo-Brézis Viability Condition for Manifold with Boundary and Corners:** The viable continuous domain $\widetilde{\mathcal{M}}_{\mathrm{flow}}$ is defined by the intersection of smooth

bounding hypersurfaces $g_i(x) \geq 0$:

$$g_1(x) = 1500x_2 - x_4 \geq 0 \quad \text{(MTU bound)} \tag{9}$$
$$g_2(x) = x_4 - x_{11} - 20x_2 \geq 0 \quad \text{(Header + payload lower bound)} \tag{10}$$
$$g_3(x) = x_{11} - x_2(20 \cdot \mathbb{I}_{x_1=6} + 8 \cdot \mathbb{I}_{x_1=17}) \geq 0 \tag{11}$$
$$g_4(x) = x_2(60 \cdot \mathbb{I}_{x_1=6} + 8 \cdot \mathbb{I}_{x_1=17}) - x_{11} \geq 0 \tag{12}$$
$$g_5(x) = x_6 - 1 \geq 0 \quad (x_{6,\min} = 1\,\mu\text{s}) \tag{13}$$
$$g_6(x) = x_2 - 1 \geq 0 \tag{14}$$
$$g_7(x) = 1.1 \times 10^8 - x_6 \geq 0 \quad \text{(flow-timeout cap)} \tag{15}$$

Because all constraint bounding functions $g_1(x), \ldots, g_7(x)$ are affine, the Bouligand contingency (tangent) cone $T_{\widetilde{\mathcal{M}}_{\text{flow}}}(x)$ at any boundary or corner point coincides exactly with the intersection of the supporting half-spaces:

$$T_{\widetilde{\mathcal{M}}_{\text{flow}}}(x) = \bigcap_{i:g_i(x)=0} \{v \in \mathbb{R}^{12} \mid \nabla g_i(x) \cdot v \geq 0\} \tag{16}$$

On any active boundary face where $g_i(x) = 0$, the inward normal velocity satisfies $\nabla g_i(x) \cdot \dot{x}(t) \geq 0$: $\nabla g_1 \cdot \dot{x} = (1500 - L_0)u_3(t) - u_2(t) \geq 0$ under admissible controls; $\nabla g_2 \cdot \dot{x} = u_2(t) \geq 0$; $\nabla g_3 \cdot \dot{x} = (H_0 - H_{\min})u_3(t) \geq 0$; $\nabla g_4 \cdot \dot{x} = (H_{\max} - H_0)u_3(t) \geq 0$; $\nabla g_5 \cdot \dot{x} = u_1(t) \geq 0$; $\nabla g_6 \cdot \dot{x} = u_3(t) \geq 0$; $\nabla g_7 \cdot \dot{x} = -u_1(t) \geq 0$ on $\{x_6 = 1.1 \times 10^8\}$, because the admissibility rule forces $u_1(t) = 0$ there. At boundary corners (intersections of multiple active faces), $\dot{x}(t) \in T_{\widetilde{\mathcal{M}}_{\text{flow}}}(x)$. By the Nagumo-Brézis viability theorem (Nagumo, 1942; Brezis, 1970), $\widetilde{\mathcal{M}}_{\text{flow}}$ is forward-invariant under the dynamical system.

- **TCP Window Trajectory:** $\dot{x}_7 = u_4(t)\mathbb{I}_{x_1=6}\,\text{sat}((W_{\text{target}} - x_7)/\epsilon_{\text{win}})$. For UDP ($x_1 = 17$), $\vec{v}_{\text{win}} \equiv \mathbf{0}$, ensuring $x_7(t) \equiv 0$. For TCP ($x_1 = 6$), $x_7(t)$ moves monotonically toward $W_{\text{target}} \in [0, 65535]$ without crossing it, so it stays between $x_7(0)$ and W_{target} and hence within $[0, 65535]$ for all $t \geq 0$.

- **Global Existence and Uniqueness:** Because denominators $x_6(t) \geq x_6(0) > 0$ and $x_2(t) \geq x_2(0) \geq 1$ are bounded away from zero, the vector fields $\sum_{k=1}^{4} u_k(t)\vec{v}_k(x)$ are Carathéodory functions with linear growth. By Carathéodory's existence and uniqueness theorem (Carathéodory, 1918), solutions exist uniquely as absolutely continuous trajectories globally on $[0, \infty)$.

□

S2. Auxiliary algorithms

Algorithm S4 gives the general per-round Lie-guided adversarial training scheme for tree ensembles. The experiments in Section 7 of the main paper run a single-round instance of it.

Algorithm S1 Adaptive Directional Lie Derivative Estimator

Require: Classifier F, current flow state x, feature schema $\mathcal{S}$
Ensure: List of candidate evaluation tuples $\mathcal{D} = \{(k, \nabla_k, \vec{v}_k, \Delta s_k, f_k)\}_{k=1}^{4}$ (with $f_k = F(x_{\text{test}})$ the probed score), total exploratory queries q

$\mathcal{D} \leftarrow \emptyset$
$q \leftarrow 0$ {Counter for candidate exploratory probes}
{1. Evaluate Temporal Dilation Generator}
$\vec{v}_1 \leftarrow \vec{v}_{\text{time}}(x)$
$\Delta s_1 \leftarrow \max(1000.0, x_6 \cdot 0.50)$ {Adaptive temporal displacement in μs}
$x_{\text{test}} \leftarrow \text{ProjectManifoldInvariants}(x + \Delta s_1 \cdot \vec{v}_1, \mathcal{S})$
$\Delta t_1 \leftarrow \Delta s_1 / 10^6$ {Convert duration step to seconds}
$f_1 \leftarrow F(x_{\text{test}})$; $\nabla_1 \leftarrow (f_1 - F(x))/\Delta t_1$ {Derivative per second of delay}
Append $(1, \nabla_1, \vec{v}_1, \Delta s_1, f_1)$ to $\mathcal{D}$
$q \leftarrow q + 1$
{2. Evaluate Payload Padding Generator}
$\vec{v}_2 \leftarrow \vec{v}_{\text{pad}}(x)$
$\Delta s_2 \leftarrow \max(50.0, x_4 \cdot 0.25)$ {Adaptive forward byte padding step}
$x_{\text{test}} \leftarrow \text{ProjectManifoldInvariants}(x + \Delta s_2 \cdot \vec{v}_2, \mathcal{S})$
$f_2 \leftarrow F(x_{\text{test}})$; $\nabla_2 \leftarrow (f_2 - F(x))/\Delta s_2$
Append $(2, \nabla_2, \vec{v}_2, \Delta s_2, f_2)$ to $\mathcal{D}$
$q \leftarrow q + 1$
{3. Evaluate Chaff Packet Injection Generator}
$\vec{v}_3 \leftarrow \vec{v}_{\text{chaff}}(x)$
$\Delta s_3 \leftarrow \max(1.0, \text{round}(x_2 \cdot 0.50))$ {Adaptive chaff packet injection count}
$x_{\text{test}} \leftarrow \text{ProjectManifoldInvariants}(x + \Delta s_3 \cdot \vec{v}_3, \mathcal{S})$
$f_3 \leftarrow F(x_{\text{test}})$; $\nabla_3 \leftarrow (f_3 - F(x))/\Delta s_3$
Append $(3, \nabla_3, \vec{v}_3, \Delta s_3, f_3)$ to $\mathcal{D}$
$q \leftarrow q + 1$
{4. Evaluate TCP Window Morphing Generator}
if $x[1] = 6$ **then**
    {TCP flows only; window size is inapplicable to UDP}
    $\vec{v}_4 \leftarrow \vec{v}_{\text{win}}(x)$
    $\Delta s_4 \leftarrow |64240.0 - x[7]|$
    if $\Delta s_4 > 0$ **then**
        $x_{\text{test}} \leftarrow \text{ProjectManifoldInvariants}(x + \Delta s_4 \cdot \vec{v}_4, \mathcal{S})$
        $f_4 \leftarrow F(x_{\text{test}})$; $\nabla_4 \leftarrow (f_4 - F(x))/(\Delta s_4/65535.0)$ {Normalized window gradient}
        Append $(4, \nabla_4, \vec{v}_4, \Delta s_4, f_4)$ to $\mathcal{D}$
        $q \leftarrow q + 1$
    end if
end if
return $\mathcal{D}, q$

S3. Per-family results

Table S1 breaks the XGBoost results of the main text down by attack family (families with at least five flows in the attacked cohort). On CSE-CIC-IDS2018 every family in the cohort is evaded in 100.0% of the flows (DoS 415 flows, DDoS 370, brute force 176, botnet 35), with 5.5 queries for DoS and 21.1 for DDoS. On CIC-IDS2017 only brute force (64.3% on 14 flows) and DoS (91.6%) fall below 100%. On UNSW-NB15 more than half of the cohort are exploits, which are evaded in only 51.0% of 520 flows and so set the aggregate figure; generic attacks are similar (51.7% on 29 flows), while fuzzers (80.1%), reconnaissance (68.8%) and

Algorithm S2 Analytical Manifold Projection Operator $\mathcal{P}_{\mathcal{M}}$

Require: Perturbed vector $x \in \mathbb{R}^{12}$, feature schema $\mathcal{S}$, optional baseline flow x_{orig}
Ensure: Projected flow vector $x_{\text{proj}} \in \mathcal{M}_{\text{flow}}$

$x_{\text{proj}} \leftarrow \text{copy}(x)$
if x_{orig} is provided **then**
  $x_{\text{proj}}[j] \leftarrow x_{\text{orig}}[j]$ for all $j \in \mathcal{I}_{\text{immut}}$ {Preserve immutable features}
  $x_{\text{proj}}[j] \leftarrow \max(x_{\text{orig}}[j], x_{\text{proj}}[j])$ for all $j \in \mathcal{I}^{+}_{\text{mono}}$ {Enforce monotonicity lower bounds}
end if
{Enforce discrete bounds and protocol-dependent packet/header sizes}
$x_{\text{proj}}[2] \leftarrow \max(1.0, \text{round}(x_{\text{proj}}[2]))$ {Integer Forward Packets}
$x_{\text{proj}}[3] \leftarrow x[3]$ {Immutable Backward Packets Preserved}
$H_{\min} \leftarrow$ **if** $x_{\text{proj}}[1] = 6$ **then** 20.0 **else** 8.0 {Min transport header: 20B TCP, 8B UDP}
$H_{\max} \leftarrow$ **if** $x_{\text{proj}}[1] = 6$ **then** 60.0 **else** 8.0 {Max transport header: 60B TCP, 8B UDP}
$x_{\text{proj}}[11] \leftarrow \min(x_{\text{proj}}[2] \times H_{\max}, \max(x_{\text{proj}}[2] \times H_{\min}, \text{round}(x_{\text{proj}}[11])))$ {Transport Header Length}
$x_{\text{proj}}[4] \leftarrow \min(x_{\text{proj}}[2] \times 1500.0, \max(0.0, \text{round}(x_{\text{proj}}[4])))$ {Forward Bytes $\in [0, 1500N_{\text{fwd}}]$}
$x_{\text{proj}}[5] \leftarrow x[5]$ {Immutable Backward Bytes Preserved}
$x_{\text{proj}}[6] \leftarrow \min(1.1 \times 10^8, \max(1.0, \text{round}(x_{\text{proj}}[6])))$ {Integer Duration $\geq 1\mu$s, capped below flow timeout}
$x_{\text{proj}}[7] \leftarrow$ **if** $x_{\text{proj}}[1] = 6$ **then** $\text{clamp}(\text{round}(x_{\text{proj}}[7]), 0.0, 65535.0)$ **else** 0.0 {TCP Window $[0, 65535]$, 0 for UDP}
{Analytically recalculate coupled non-linear flow rates}
$T_{\text{sec}} \leftarrow x_{\text{proj}}[6]/10^6$
$B_{\text{tot}} \leftarrow x_{\text{proj}}[4] + x_{\text{proj}}[5]$
$N_{\text{tot}} \leftarrow x_{\text{proj}}[2] + x_{\text{proj}}[3]$
$x_{\text{proj}}[8] \leftarrow B_{\text{tot}}/T_{\text{sec}}$ {Flow Bytes/s}
$x_{\text{proj}}[9] \leftarrow N_{\text{tot}}/T_{\text{sec}}$ {Flow Packets/s}
$x_{\text{proj}}[10] \leftarrow x_{\text{proj}}[4]/x_{\text{proj}}[2]$ {Forward Packet Length Mean}
return x_{proj}

DoS (65.7%) are easier. Families with fewer than five cohort flows are not reported.

S4. Ablations

The ablations use the CUDA XGBoost target of CSE-CIC-IDS2018 and a balanced cohort of $N = 300$ detected malicious flows (up to 50 per attack family). Table S2 reports three sweeps: Panel A varies the step budget M, Panel B isolates generator subsets, and Panel C compares operational cost regimes. The TSBS relative snapping offset ϵ_{rel} was swept over $\{0, 10^{-6}, 10^{-4}, 10^{-2}, 1\}$.

Step budget. ASR rises from 52.7% at $M = 1$ (5.0 queries) through 89.7% at $M = 3$ and 96.3% at $M = 5$ to 100.0% at $M = 20$ (10.3 queries) and is flat beyond. A single step already evades more than half of the cohort, which is why the mean query count stays near ten. Physical feasibility is 100.0% for every budget.

Generators. No single generator suffices. Temporal dilation alone evades 57.0% of the flows but needs 33.5 queries and adds 39.8 s of flow duration on average; window morphing alone evades 53.0% with 2.6 queries and no added delay; chaff alone 47.7%; padding alone 30.0%. Pairs are better than singles (time plus chaff 82.7%, time plus padding 68.0%, padding plus chaff 49.0% with no added delay), and the complete cone reaches 100.0% with 10.3 queries and 0.22 s of mean delay. The full cone needs fewer queries than the time-based subsets because the cheap window and chaff moves are available when delay alone would

Algorithm S3 TSBS Candidate Generation (GenerateCandidates)

Require: Current flow $x^* \in \mathcal{M}_{\text{flow}}$, active splits active_splits (pairs (j, θ_j)), tolerance ϵ_{rel}
Ensure: Candidate boundary-crossing pool $\mathcal{C}$ (before manifold projection)

$\mathcal{C} \leftarrow \emptyset$
{1. Monotonic features: closest split strictly above $x^*[j]$; UDP header x_{11} is immutable at 8B}
for each $j \in \mathcal{I}^+_{\text{mono}}$ **with** NOT $(j = 11 \wedge x^*[1] = 17)$ **do**
  $\theta^*_j \leftarrow \min\{\theta_j \mid (j, \theta_j) \in \text{active_splits} \ \wedge \ \theta_j > x^*[j]\}$
  if θ^*_j exists **then**
    $\epsilon_j \leftarrow 1.0$ **if** $j \in \{2, 4, 11\}$ **else** $\text{round}(\epsilon_{\text{rel}} \cdot \max(1.0, |\theta^*_j|))$
    $x_c \leftarrow \text{copy}(x^*)$; $x_c[j] \leftarrow \theta^*_j + \epsilon_j$
    if $j = 11$ **then**
      $x_c[11] \leftarrow \min(60.0 \cdot x_c[2], x_c[11])$
      $x_c[4] \leftarrow \min(1500.0 \cdot x_c[2], \max(0.0, x_c[4]))$
    end if
    $\mathcal{C} \leftarrow \mathcal{C} \cup \{x_c\}$
  end if
end for
{2. Invert coupled rate splits (x_8, x_9) into duration dilation}
for each $(r, a, b) \in \{(8, x^*[4], x^*[5]), (9, x^*[2], x^*[3])\}$ **do**
  $\theta^*_r \leftarrow \max\{\theta_r \mid (r, \theta_r) \in \text{active_splits} \ \wedge \ 0 < \theta_r < x^*[r]\}$
  if θ^*_r exists **and** $d \leftarrow 10^6(a + b)/\theta^*_r + 10.0$ satisfies $d \leq 1.1 \times 10^8$ **then**
    $x_c \leftarrow \text{copy}(x^*)$; $x_c[6] \leftarrow d$; $\mathcal{C} \leftarrow \mathcal{C} \cup \{x_c\}$
  end if
end for
{3. Invert coupled packet length splits (x_{10}) into chaff packet injection}
$H_0 \leftarrow 32.0$ **if** $(x^*[1] = 6 \wedge x^*[11]/x^*[2] \geq 26.0)$ **else** 20.0 **if** $x^*[1] = 6$ **else** 8.0
$L_0 \leftarrow 20.0 + H_0$ {52B/40B TCP (with/without RFC 7323 options), 28B UDP}
$\theta^*_{10} \leftarrow \max\{\theta_{10} \mid (10, \theta_{10}) \in \text{active_splits} \ \wedge \ L_0 < \theta_{10} < x^*[10]\}$
if θ^*_{10} exists **then**
  $N \leftarrow \lfloor (x^*[4] - \theta^*_{10} x^*[2])/(\theta^*_{10} - L_0) \rfloor + 1$
  if $N \leq 1000$ **then**
    {Chaff budget cap: $N \leq 1000$ packets}
    $x_c \leftarrow \text{copy}(x^*)$; $x_c[2] \leftarrow x^*[2] + N$
    $x_c[4] \leftarrow x^*[4] + N \cdot L_0$; $x_c[11] \leftarrow x^*[11] + N \cdot H_0$
    $\mathcal{C} \leftarrow \mathcal{C} \cup \{x_c\}$
  end if
end if
{4. TCP window morphing to benign OS baseline}
if $x^*[1] = 6 \ \wedge \ x^*[7] \neq 64240.0$ **then**
  $x_c \leftarrow \text{copy}(x^*)$; $x_c[7] \leftarrow 64240.0$; $\mathcal{C} \leftarrow \mathcal{C} \cup \{x_c\}$
end if
return $\mathcal{C}$

stall.

Cost regimes and snapping offset. These do not discriminate on this corpus. All four cost regimes reach 100.0% ASR with 9.8–13.9 queries, and the snapping offset leaves ASR at 100.0% and the queries between 10.2 and 10.3. The regimes do change which generator dominates (time under the balanced regime, window morphing under the latency-sensitive regime) and the mean added delay (216 ms under regime A against 91 ms under regime B), so FFE responds to the operator's cost model, but because the corpus is saturated we have no evidence on how the regime affects success; a harder target such as UNSW-NB15 would

Algorithm S4 Lie-Guided Adversarial Training (LGAT) for Tree Ensembles

Require: Training set $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N}$, number of boosting rounds T, adversarial ratio $\beta = 0.5$, step budget $M_{\text{train}} = 5$
Ensure: Robust tree ensemble $F_T(x)$
Initialize base tree $F_0 \leftarrow \text{FitCleanBaseTree}(\mathcal{D}_{\text{train}})$
for $t = 1$ **to** T **do**
  $\mathcal{D}_{\text{aug}} \leftarrow \emptyset$
  for each $(x_i, y_i) \in \mathcal{D}_{\text{train}}$ **do**
    if $y_i = 1$ **and** $\text{Uniform}(0, 1) < \beta$ **then**
      $x_{\text{adv}} \leftarrow \text{FFE}(F_{t-1}, x_i, \text{budget} = M_{\text{train}})$
      Add $(x_{\text{adv}}, 1)$ to $\mathcal{D}_{\text{aug}}$
    else
      Add (x_i, y_i) to $\mathcal{D}_{\text{aug}}$
    end if
  end for
  Compute negative gradients g_i and Hessians h_i on $\mathcal{D}_{\text{aug}}$
  Fit tree f_t using robust split scoring $\text{Gain}_{\text{robust}}$
  $F_t(x) \leftarrow F_{t-1}(x) + \eta \cdot f_t(x)$
end for
return F_T

Table 1: *

Table S1. FFE against XGBoost by attack family.

Dataset	**Attack family**	n	**Real ASR (%)**	**Queries**
CSE-CIC-IDS2018	DoS	415	100.0	5.5
	DDoS	370	100.0	21.1
	BruteForce	176	100.0	13.9
	Botnet	35	100.0	13.1
CIC-IDS2017	DoS	428	91.6	19.8
	PortScan	244	100.0	19.3
	DDoS	238	100.0	5.0
	InfiltrationBeacon	71	100.0	18.8
	BruteForce	14	64.3	71.1
	Botnet	5	100.0	11.0
UNSW-NB15	Exploits	520	51.0	112.1
	Fuzzers	221	80.1	57.4
	Reconnaissance	138	68.8	75.7
	DoS	70	65.7	78.8
	Generic	29	51.7	106.3
	Shellcode	11	100.0	9.4
	Backdoor	8	75.0	72.0

be needed for that.

Table 2: *

Table S2. Ablation of FFE on CUDA XGBoost, CSE-CIC-IDS2018 ($N = 300$). Delay is the mean added flow duration.

Configuration	**ASR (%)**	**PFR (%)**	**Queries**	**Added delay (s)**
Panel A: step budget M				
$M = 1$	52.7	100.0	5.0	0.09
$M = 2$	61.0	100.0	6.7	0.13
$M = 3$	89.7	100.0	8.2	0.20
$M = 5$	96.3	100.0	9.2	0.20
$M = 10$	99.7	100.0	10.2	0.21
$M = 20$	100.0	100.0	10.3	0.22
$M = 50$	100.0	100.0	10.3	0.22
Panel B: generator subset ($M = 20$)				
Temporal Dilation Only	57.0	100.0	33.5	39.76
Payload Padding Only	30.0	100.0	4.1	0.00
Chaff Injection Only	47.7	100.0	7.6	0.00
Window Morphing Only	53.0	100.0	2.6	0.00
Time + Padding	68.0	100.0	40.6	27.71
Time + Chaff	82.7	100.0	33.2	18.11
Padding + Chaff	49.0	100.0	11.0	0.00
Complete Lie Cone	100.0	100.0	10.3	0.22
Panel C: operational cost regime ($M = 20$)				
A (Default / Balanced)	100.0	100.0	10.3	0.22
B (Latency-Sensitive)	100.0	100.0	9.8	0.09
C (Unconstrained Delay)	100.0	100.0	13.9	0.24
D (Uniform / Flat)	100.0	100.0	13.7	0.13

S5. Cross-architecture transferability

Flows crafted by FFE against a source ensemble were scored by three independent target architectures ($N = 300$ detected CSE-CIC-IDS2018 flows; Table S3). Flows crafted against XGBoost evade LightGBM in 69.0% of the cases, Random Forest in 81.3% and CatBoost in 37.7%. LightGBM flows evade XGBoost in 82.3% and Random Forest in 83.0%, CatBoost flows evade LightGBM in 66.7%, and Random Forest flows evade XGBoost in 79.0%. CatBoost is the hardest target for every source (34.0%–37.7%); this is consistent with its symmetric oblivious trees placing thresholds differently from the leaf-wise and depth-wise ensembles, although we did not test that explanation. Attacks against CatBoost itself succeed in 99.3% of the cases, so its resistance to transfer reflects dissimilar boundaries and not a stronger model. Excluding CatBoost as target, 54%–83% of the transferred flows evade.

Table 3: *

Table S3. Transferability of FFE flows (attack success rate in %, $N = 300$; diagonal: attack against the source model itself).

Source \ Target	**XGBoost (CUDA)**	**LightGBM**	**CatBoost**	**Random Forest**
XGBoost (CUDA)	**100.0**	69.0	37.7	81.3
LightGBM	82.3	**100.0**	34.0	83.0
CatBoost	54.3	66.7	**99.3**	63.7
Random Forest	79.0	64.0	37.0	**100.0**